



**Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-05.
Department of MCA**

Course Detail Document

Academic Year: 2025-26

Course: SOFTWARE TESTING AND QUALITY ASSURANCE

Term: I

Class: S.Y.

Division: A&B

Name of the Course Instructor

Dr.Shivani Budhkar

Dr.Mrs.R.S.Bansode

Name of the HOD

Dr.Shivani Budhkar



Vision of the Institute

- Creation of a collaborative academic environment to foster professional excellence and ethical values.

Mission of the Institute

- To develop outstanding professionals with high ethical standards capable of creating and managing global enterprises.
- To foster innovation and research by providing a stimulating learning environment.
- To ensure equitable development of students of all ability levels and backgrounds.
- To be responsive to changes in technology, socio-economic and environmental conditions.
- To foster and maintain mutually beneficial partnerships with alumni and industry.

Objectives of the Institute

- To develop infrastructure appropriate for delivering quality education
- To develop the overall personality of students who will be innovators and future leaders capable of prospering in their work environment.
- To inculcate ethical standards and make students aware of their social responsibilities.
- Promote close interaction among industry, faculty and students to enrich the learning process and enhance career opportunities.
- Encourage faculty in continuous professional growth through quality enhancement programs and research and development activities.
- Foster a healthy work environment which allows for freedom of expression and protection of the rights of all stakeholders through open channels of communication



Vision of the Department

“To develop Competent Technocrats in the field of Computer applications imbued with human values”

Mission of the Department

- M1. To impart knowledge in the field of Computer applications with a focus on developing the required competencies.
- M2. To improve technical skill of the students through practical and hands-on experience.
- M3. To enhance the quality of the students by collaboration with Alumni and Industry.
- M4. To make students socially responsible citizens.

Program Educational Objectives (PEOs)

- PEO 1: Graduates will possess the broad knowledge of computer applications for successful careers in industry.
- PEO 2: Graduates will exhibit professionalism, ethical attitude, communication skills, team work in their profession and adapt to current trends by engaging in lifelong learning.
- PEO 3: Graduates will contribute as responsible citizens with a commitment to the sustainable development of society.



Program Outcomes (POs)

At the end of both semesters students' will be able to–

PO1: Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

PO2: Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.

PO3: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.

PO4: Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO5: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

PO6: Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

PO7: Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware

PO8: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.



Course Structure



Progressive Education Society's
 MODERN COLLEGE OF ENGINEERING, Pune -05.
 (An Autonomous Institute Affiliated to Savitribai Phule Pune University)
 Department of Master of Computer Applications

S.Y. MCA Curriculum Structure (w.e.f. 2025-2026) (2024 Pattern)

	Semester- III															
Course Type	Course Code	Course Title	Teaching Scheme (Hours / Week)				Examination / Assessment Scheme and Marks						Credit Scheme			
			L	T	P	SL	CIE	SEE	TW	PR	OR	Total	L	T	P	Total
MJ	MCA01601	Software Testing And Quality Assurance	3	-	-	-	40	60	-	-	-	100	3	-	-	3
MJ	MCA01602	Data Science	3	-	-	-	40	60	-	-	-	100	3	-	-	3
MJ	MCA01603	Web Programmmg	3	-	-	1	40	60	-	-	-	100	3	-	-	3
PE	MCA10601A/B/C	Elective –III	3	-	-	-	40	60	-	-	-	100	3	-	-	3
MJ	MCA01604	Data Science Laboratory	-	-	4	-	-	-	25	25	-	50	-	-	2	2
MJ	MCA01605	Web Programming Laboratory	-	-	4	-	-	-	25	25	-	50	-	-	2	2
PE	MCA10602	Laboratory Practice- III	-	-	2	-	-	-	25	-	-	25	-	-	1	1
RP	MCA08601	Research Project	-	-	6	-	-	-	25	-	50	75	-	-	3	3
SE	SDV05601	Skill Development-I	-	2	-	-	-	-	50	-	-	50	-	2	-	2
Total Hours/week, Marks, Credits			12	2	16	1	160	240	150	50	50	650	12	2	8	22
TOTAL			30 Hrs/week				400 Marks (TH)		250 Marks (PR)							

Elective – III:

1. MCA10601A Deep Learning
2. MCA10601B Cloud Computing
3. MCA10601C Information retrieval



K. Joshi
 30/04/2025.
 (Prof. Dr. Mrs. Kalyan R. Joshi)
 Chairperson Academic Council, PESMCOE

S. Itkar
 (Prof. Dr. Mrs. Suhasini A. Itkar)
 Chairperson BOS



COURSE DETAILS DOCUMENT
Academic Year: 2025-2026 (TERM– I)

Class: SYMCA

Course Name: SOFTWARE TESTING AND QUALITY ASSURANCE

Course Code: MCA01601

Course No.: CO601

Teaching Scheme

Marking Scheme

Theory –3 Hrs/week

Theory Marks

CIE: 40 SEE: 60

Prerequisites of the Course

Basic knowledge of Software Development Methodologies

Companion Course (supporting practical course, if any)

Laboratory Practice- III

Course Objectives

To design and implement effective test cases for software applications using selenium.

- To apply deep learning techniques in solving real world problems.
- To create and manage cloud-based database services, perform basic operations, and configure load balancers for efficient traffic distribution.
- To understand the information retrieval process using standards available tools.

CO No.	Year of study 2024-25	Mapping to POs		
		Substantial	Moderate	Low
C601.1	Illustrate basics of software quality assurance, and quality control.	PO1,PO2	-	-
C601.2	Design test plan, test cases and defect repository using real world applications.	PO1,PO2,PO8	-	-
C601.3	Apply the concept of white box and block box testing techniques.	PO1,PO2,PO3,PO8	-	-
C601.4	Explain various software testing types.	PO1,PO2,PO3	-	-
C601.5	Summarize software test automation concepts.	PO1,PO2,PO4	-	-
C601.6	Apply software testing automation concepts using open source Selenium.	PO1,PO2,PO4,PO8	-	-

*Mapping details based on CO-PO mapping.



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CO to PO mapping

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO601.1	3	3	-	-	-	-	-	-
CO601.2	3	3	3	-	-	-	-	3
CO601.3	3	3	3	-	-	-	-	3
CO601.4	3	3	-	-	-	-	-	-
CO601.5	3	3	-	3	-	-	-	-
CO601.6	3	3	-	3	-	-	-	3
CO601	3	3	3	3	-	-	-	3

Note: In the table above, mapping strength is defined as:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) ‘-’: It there is no correlation

Justification of CO to PO Mapping

CO	PO	Strength	Justification
C601.1	PO1	3	Substantially mapped as the students will be able to apply knowledge to illustrate different approaches of quality management, assurance, and quality standard to software system.
	PO2	3	To ensure software meets customer requirements, students must critically evaluate the product against specified standards and user expectations.
C602.2	PO1	3	Substantially mapped as the students will be able to apply knowledge to create test plan, test cases and defect repository
	PO2	3	Substantially mapped as the students will be able to analyze real world applications to create test plan, test cases and defect repository
	PO3	3	Substantially mapped as the students will be able to design solutions of real world applications such as create test plan, test cases and defect repository
	PO8	3	Designing test plans, test cases, and defect repositories using real-world applications helps students develop change management skills by adapting to evolving technologies and workflows. It fosters continuous learning and effective handling of software changes, aligning well with PO8.
C602.3	PO1	3	Substantially mapped as the students will be able to apply knowledge of white box and block box testing techniques for real world applications
	PO2	3	Substantially mapped as the students will be able to analyze real world applications and choose white box and block box testing techniques for testing particular application
	PO3	3	Substantially mapped as the students will be able to design testing solutions in real world applications by designing white box test cases and block box test cases
	PO8	3	Applying white box and black box testing techniques requires students to continuously update their skills with evolving testing methods and tools, promoting adaptability and lifelong learning. This aligns well with PO8's focus on change management and keeping up with contemporary technologies.
C603.4	PO1	3	Substantially mapped as the students will be able to apply knowledge of software testing types to choose appropriate testing type for real world applications



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	PO2	3	Substantially mapped as the students will be able to analyze real world software applications under testing and select testing types to test software applications
C603.5	PO1	3	Substantially mapped as the students will be able to apply knowledge automation tool to choose automation tool for software testing
	PO2	3	Substantially mapped as the students will be able to analyze real world applications under test and choose automation tool for software testing
	PO4	3	Summarizing software test automation concepts involves selecting and applying modern tools and techniques to improve testing efficiency, directly aligning with PO4's emphasis on using computational tools while understanding their capabilities and limitations.
C603.6	PO1	3	Substantially mapped as the students will be able to apply knowledge of software testing automation concepts for use of Selenium tool
	PO2	3	Substantially mapped as the students will be able to analyze real world applications under test and apply software testing automation concepts using Selenium
	PO4	3	Substantially mapped as the students will be able to design testing solutions in real world applications by using Selenium automation tool.
	PO8	3	Applying software testing automation using Selenium requires students to learn and adapt to current industry tools, fostering their ability to keep up with evolving technologies and change management practices, which aligns well with PO8.



UNIT WISE SYLLABUS

Course Contents	
Unit I	Introduction to Software Quality Assurance
Software Quality Assurance: Quality, quality assurance, quality control, metrics in software development, quality management system, quality standards- ISO 9000 series standards, capability maturity model, criteria for the success of a software project, process-oriented software development.	
Unit II	Basics of Software Testing
Introduction of Software Testing: Definition, objectives of testing, verification and validation, levels of testing, testing approaches, testing life cycle, software testing principles, criteria for completion of testing, agile testing methodologies -Test-Driven Development(TDD), Behavior-Driven Development(BDD), and Acceptance Test-Driven Development(ATDD). Test Plan and Test Cases: Test plan preparation, management and execution of test plan. Test case-definition, designing of test cases, case studies. Defect management: Origins of defects, classification of defects, bug life cycle, defect repository, case studies.	
Unit III	Software Testing Techniques
White-box testing methodologies: Need of white box testing, unit testing, integration testing, code coverage testing, basis path testing, mutation testing, challenges in white box testing, examples. Black-box testing methodologies: Need of black box testing, boundary value analysis, equivalence partitioning, decision table testing, state transition testing, Case studies.	
Unit IV	Software Testing Strategies
Testing Strategies: Psychology of testing, types of testing, system and acceptance testing, functional and non-functional testing, performance testing, regression testing, ad-hoc testing.	
Unit V	Software Testing Tools : An Overview
Introduction to Automation testing: Need for automation testing tools, functional /regression testing tools, performance testing tools, test management tools. Automation Tools: Cypress, JMeter, TestNG.	
Unit VI	Selenium Tool
Selenium Tool: Introduction of selenium, brief history of the selenium project, selenium tool suite, selenium IDE, selenium RC, selenium WebDriver, selenium Grid,TestNG.	
Learning Resources:	
Text Books: 1. Srinivasan Desikan, Gopalswamy Ramesh,“Software Testing Principles and Practices”, Pearson, ISBN-10. 9788177581218. 2. David Burns,“Selenium 1.0 Testing Tools”, PACKT Open Source, ISBN-10. 1849510261.	
Reference Books: 1. Ron Patton, “Software Testing”, Pearson, Second edition, ISBN:9788177580310. 2. Rajani & Oak, “Software Testing: Methodology, Tools and Processes”, Tata McGraw-Hill, 2007, ISBN-10: 0070583528. 3. Rahul Shende,” Software Automation Testing Tools for Beginners”, Shroff Publishers and Distributors, 2012, ISBN-10. 1619030454 4. Dr.K.V.K.K Prasad , “Software Testing Tools”, Dreamtech Press, ISBN-13. 978-8177225327.	
e-Books: 1. Selenium 1.0 Testing Tool beginners guide by David Burns, ISBN: 1849510261, ISBN 13: 9781849510264 2. Burnstein, “Practical Software Testing”, Springer International Edition, ISBN 81- 8128-089-X	



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MOOC Courses: <web links>

1. <https://www.my-mooc.com/en/mooc/software-testing-fundamentals/>
2. <https://nptel.ac.in/courses/106/105/106105150/>
3. https://onlinecourses.nptel.ac.in/noc19_cs71/preview

Internal Assessment Tools (20% Weightage in Course Outcome Attainment)

Sr. No.	Assessment Tool	Marks
1	CIE-1 (Subjective Test)	20
2	CIE-2(Case Studies)	20
3	CIE-III (MCQ)	20
	Total	60
	CONVERTED TO	40

External Assessment Tools (80% Weightage in Course Outcome Attainment)

Sr. No.	Assessment Tool	Marks
1	Semester-End Examination (SEE)	60
	Total	60



Content beyond syllabus Details – Theory

Sr. No.	Content beyond syllabus (Theory-Topic)	Type of Activity (Lecture/ Seminar / Webinar / Workshop / Hands – on Training)	Name of the Resource Person	Name of Organization and Designation	Cos Mapped	POs Mapped
1	Graph based Testing	Lecture and Practical	Dr.Rama Bansode/	MCOE	CO2	PO3
2.	Cyclomatic Complexity	Lecture	Dr. Shivani Budhkar/ Manisha Patil	MCOE	CO3	PO1

Co-curricular Activity Details - Theory

Sr. No.	Name of Activity (Theory-Topic)	Type of Activity (Seminar / Webinar / Workshop / Hands – on Training)	Name of the Resource Person	Name of Organization and Designation	COs Mapped	POs Mapped
1	session on Selenium Web driver	Hands-on	Mr. Chetan Bendre	Incedo Solutions, Pune Automation Test Engineer	CO6	PO1,PO3



Department of MCA

Department of Master of Computer Applications

Curriculum Booklet - Theory Course

Class: SYMCA

Name of the Course – Software Testing and Quality Assurance

2024 Pattern

(With effect from 2025-26)

Course in charges

Dr.Shivani Budhkar
Dr.Mrs.R.S.Bansode

Module Coordinator

Dr.Shivani Budhkar

HOD

Dr.Shivani Budhkar



Teaching Plan

Sr. No.	Unit	Topics to be covered	Book Referred	Total Lecture Planned
1	I	Introduction to Software Quality Assurance	T1	7
2	II	Basics of Software Testing	T1,R1	7
3	III	Software Testing Techniques	T1	7
4	IV	Software Testing Strategies	T1	7
5	V	Software Test Automation	T1	7
6	VI	Selenium Tool	T2,T4, R2	7

Text Books:

1. Srinivasan Desikan, Gopalswamy Ramesh, "Software Testing Principles and Practices", Pearson, ISBN-10. 9788177581218.

2. David Burns, "Selenium 1.0 Testing Tools", PACKT Open Source, ISBN-10. 1849510261.

Reference Books:

1. Ron Patton, "Software Testing", Pearson, Second edition, ISBN:9788177580310.

2. Rajani & Oak, "Software Testing: Methodology, Tools and Processes", Tata McGraw-Hill, 2007, ISBN-10: 0070583528.

3. Rahul Shende, "Software Automation Testing Tools for Beginners", Shroff Publishers and Distributors, 2012, ISBN-10. 1619030454

4. Dr.K.V.K.K Prasad, "Software Testing Tools", Dreamtech Press, ISBN-13. 978-8177225327



Reference Web Links/ Research Paper/ Referred Book other than Mention in Syllabus:

- <https://doc.lagout.org/Alfred%20V.%20Aho%20-%20Data%20Structures%20and%20Algorithms.pdf>
- <https://www.cs.bham.ac.uk/~jxb/DSA/dsa.pdf>

MOOC Courses: <web links>

1. <https://www.my-mooc.com/en/mooc/software-testing-fundamentals/>
2. <https://nptel.ac.in/courses/106/105/106105150/>
3. https://onlinecourses.nptel.ac.in/noc19_cs71/preview



Unit No.-I- Fundamentals of Software Quality Assurance

Lecture No.	Details of the Topic to be covered	References
1	Software Quality Assurance: Quality, quality assurance, quality control	T1,R1
2	metrics in software development,	R1
3	quality management system	R1
4	quality standards- ISO 9000 series standards	R1
5	capability maturity model	R1
6	criteria for the success of a software project	R4
7	process-oriented software development	R4

Question Bank: Theory & Numerical Mapped to Course Outcome:

Q. 1	What is Quality Assurance?
Q. 2	What is Quality Control?
Q. 3	Differentiate between Quality Assurance and Quality Control
Q. 4	Which are criteria for the success of a software project?
Q. 5	Explain CMM/CMMI.
Q. 6	What are the Activities of SQA
Q. 7	Explain ISO900 principles
Q. 8	What is process-oriented software development



Unit No.-II- Essentials of Software Testing

Lecture No.	Details of the Topic to be covered	References
1	Introduction of Software Testing: Definition, objectives of testing, verification and validation, levels of testing	R1
2	testing approaches, testing life cycle, software testing principles, criteria for completion of testing,	R1
3	Agile testing methodologies -Test-Driven Development(TDD),Behavior-Driven Development(BDD), and Acceptance Test-Driven Development(ATDD).	https://www.browserstack.com/guide/tdd-vs-bdd-vs-atdd
4	Test Plan and Test Cases: Test plan preparation, management and execution of test plan.	R1
5	Test case- definition, designing of test cases, case studies.	R1
6	Defect management: Origins of defects, classification of defects	R1
7	bug life cycle, defect repository, case studies	https://bugbug.io/blog/software-testing/bug-life-cycle/

Question Bank: Theory & Numerical Mapped to Course Outcome:

Q. 1	Differentiate between Verification and Validation
Q. 2	What is Testing? Write down objective of Testing
Q. 3	Explain Testing Life cycle
Q. 4	What is Test plan? Prepare sample test plan.
Q. 5	What is Test plan? How to execute test plan
Q. 6	Define the term test plan, Test cases.
Q. 7	With the help of case studies write down Test plan and Test cases
Q. 8	Write note on – TDD,BDD,ATDD



Unit No.-III- Software Testing Techniques

Lecture No.	Details of the Topic to be covered	References
1	White-box testing methodologies: Need of white box testing	T1,R1
2	unit testing, integration testing	T1, R1
3	code coverage testing, basis path testing	T1, R1
4	mutation testing, challenges in white box testing, examples	T1, R1
5	Black-box testing methodologies: Need of black box testing, boundary value analysis, equivalence partitioning	T1, R1
6	decision table testing, state transition testing	T1, R1
7	Black box testing - Case studies	T1, R1

Mapped to Course Outcome:

Q. 1	Explain White Box testing methodologies
Q. 2	Explain Black Box testing methodologies
Q. 3	Explain Code coverage Testing
Q. 4	Explain Equivalence Partitioning
Q. 5	Explain Boundary value analysis with example
Q. 6	Explain decision table testing with example
Q. 7	Explain mutation testing
Q. 8	What is State Transition Testing



Unit No.-IV- Testing Strategies

Lecture No.	Details of the Topic to be covered	References
1	Testing Strategies: Psychology of testing, types of testing,	T1
2	System testing	T1
3	acceptance testing	T1
4	Functional testing	T1
5	non-functional testing, performance testing	T1
6	regression testing	T1
7	ad-hoc testing	T1

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	What is Integration testing? Explain types of it.
Q. 2	Explain System testing
Q. 3	Explain Acceptance testing
Q. 4	Explain Regression testing,
Q. 5	Explain Ad hoc Testing
Q. 6	Explain GUI Testing
Q. 7	What are different types of testing
Q. 8	Explain functional & non functional testing



Unit No.-V Software Test Automation

Lecture No.	Details of the Topic to be covered	References
1	Introduction to Automation testing: Need for automation testing tools,	R4
2	Functional testing tools	R4
3	regression testing tools	R4
4	performance testing tools	R4
5	test management tools.	R4
6	Automation Tools: Cypress, JMeter,	https://www.browserstack.com/guide/automation-testing-open-source-tools
7	Automation Tools: TestNG.	

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	What is automation testing and why is it important in software development?
Q. 2	Explain the need for automation testing tools over manual testing.
Q. 3	Discuss the advantages and challenges of using automation testing tools.
Q. 4	What is Cypress and what are its key features in automation testing?
Q. 5	Describe the role of TestNG in test automation and its key functionalities.
Q. 6	What types of tests can be automated using Cypress and JMeter respectively?
Q. 7	How does automation testing improve the efficiency of regression testing?
Q. 8	Discuss the criteria for selecting the appropriate automation tool for a given project.



Unit No.-VI- Selenium Tool

Lecture No.	Details of the Topic to be covered	References
1	Selenium Tool: Introduction of selenium, brief history of the selenium project	T2
2	selenium tool suite	T2
3	selenium IDE	T2
4	selenium RC	T2
5	selenium WebDriver	T2
6	selenium Grid	T2
7	TestNG	T2

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	Explain WebDriver
Q. 2	Explain Selenium Grid
Q. 3	Explain Software Test Automation through Selenium open source tool
Q. 4	What is scripting language? Explain it
Q. 5	Explain Selenium IDE
Q. 6	Explain Selenium RC
Q. 7	Explain brief history of Selenium
Q. 8	What are advantages of Selenium Web driver